Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 26
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

A developer is creating an application that will be deployed on IoT devices. The application will send data to a RESTful API that is deployed as an AWS Lambda function. The application will assign each API request a unique identifier. The volume of API requests from the application can randomly increase at any given time of day.
During periods of request throttling, the application might need to retry requests. The API must be able to handle duplicate requests without inconsistencies or data loss.
Which solution will meet these requirements?

A. Create an Amazon RDS for MySQL DB instance. Store the unique identifier for each request in a database table. Modify the Lambda function to check the table for the identifier before processing the request.

B. Create an Amazon DynamoDB table. Store the unique identifier for each request in the table. Modify the Lambda function to check the table for the identifier before processing the request.

C. Create an Amazon DynamoDB table. Store the unique identifier for each request in the table. Modify the Lambda function to return a client error response when the function receives a duplicate request.

D. Create an Amazon ElastiCache for Memcached instance. Store the unique identifier for each request in the cache. Modify the Lambda function to check the cache for the identifier before processing the request.

Show Suggested Answer

by Untamables at *March 23, 2023, 11:48 a.m.*

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Comments

Devon_Fazekas Highly Voted 2 years, 10 months ago

Selected Answer: B

I originally thought ElastiCache would provide the sufficient session management of the unique identifiers with the least latency. But apparently, the scope of this question revolves around durability, not latency. Hence, a persistent storage is better suited. And while RDS is a viable solution for durability and performance, the question specifies IoT devices which typically produce unstructured data that is better handled by No-SQL services like DynamoDB.

upvoted 33 times

Untamables Highly Voted 3 years ago

Selected Answer: B

B
The resolution is to make the Lambda function idempotent.
https://repost.aws/knowledge-center/lambda-function-idempotent
https://aws.amazon.com/builders-library/making-retries-safe-with-idempotent-APIs/

upvoted 9 times

gordilloedwin Most Recent 10 months, 2 weeks ago

Selected Answer: B

Yeah, I thought of C as well... but the cache entries will be evicted sooner rather than later, and thus the new entry will be processed again. Cache is just temporary

upvoted 1 times

sumanshu 1 year, 3 months ago

Selected Answer: B

A) Eliminated - RDS is not as scalable as DynamoDB for handling unpredictable spikes in traffic.

D) Eliminated - Memcached is an in-memory cache that is not designed for persistence or durability. If the instance is restarted or the data is evicted due to memory constraints, the identifiers could be lost

upvoted 4 times

sumanshu 1 year, 3 months ago

C) Eliminated - Returning a client error response for duplicate requests is not a good approach

upvoted 2 times

trieudo 1 year, 3 months ago

Selected Answer: B

keyword:
- AWS Lambda function
- increase at any given time
- retry requests
- handle duplicate requests

==> Discard A, D: violate 'increase at any given time', it is not automatically, because it is not fit with serverless service 'AWS Lambda function'
==> Discard C: violate 'retry requests + handle duplicate requests', => different response from success vs. error error, make it is non-idempotency
=> Discard D: violate 'inconsistencies or data loss.' it doest not store persistent data.

B is best solution

upvoted 3 times

f271c23 1 year, 3 months ago

Selected Answer: C

I think C has the merits to be a right answer. It has more specific handling to duplicate requests . The answer B does the same but the langugage is more explicite in option C

upvoted 1 times

ElFaramawi 1 year, 10 months ago

Why not C?
According to the question "During periods of request throttling, the application might need to retry requests", this indicate that lambda should returns client error, so the application can make another retry request to fix the problem

upvoted 3 times

cocolavayen 1 year, 9 months ago

same doubt

upvoted 1 times

queekao 1 year, 8 months ago

because c don't mention about retry actions

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: B

B is the correct answer.

upvoted 1 times

Abdullah22 2 years ago

Selected Answer: B

Consistency: Memcached does not provide built-in support for atomic operations or conditional writes like DynamoDB does. Handling duplicate requests and ensuring consistency would require additional application logic and complexity.

upvoted 1 times

Tony88 2 years, 7 months ago

Selected Answer: B

Cache topic.
So Elastic Redis and DynamoDB both can be used as a cache solution.
If you want high performance, low latency, go with Redis
If you want persistent storage, go with DyanmoDB.

upvoted 5 times